EMG Signal Decomposition Using Motor Unit Potential Train Validity

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***Abstract*—A system to resolve an intramuscular electromyographic (EMG) signal into its component motor unit potential trains (MUPTs) is presented. The system is intended mainly for clinical applications where several physiological parameters of motor units (MUs), such as their motor unit potential (MUP) templates and mean firing rates, are of interest. The system filters an EMG signal, detects MUPs, and clusters and classifies the detected MUPs into MUPTs. Clustering is partially based on the K-means algorithm, and the supervised classification is implemented using a certainty-based algorithm. Both clustering and supervised classification algorithms use MUP shape and MU firing pattern information along with signal dependent assignment criteria to obtain robust performance across a variety of EMG signals. During classification, the validity of extracted MUPTs are determined using several supervised classifiers; invalid trains are corrected and the assignment threshold for each train is adjusted based on the estimated validity (i.e., adaptive classification). Performance of the developed system in terms of accuracy (A_c), assignment rate (A_r), correct classification rate (CC_r), and the error in estimating the number of MUPTs represented in the set of detected MUPs (E_{NMUPTs}) was evaluated using 32 simulated and 30 real EMG signals comprised of 3–11 and 3–15 MUPTs, respectively. The developed system, with average CC_r of 86.4% for simulated and 96.4% for real data, outperformed a previously developed EMG decomposition system, with average CC_r of 71.6% and 89.7% for simulated and real data, by 14.7% and 6.7%, respectively. In terms of E_{NMUPTs}, the new system, with average E_{NMUPTs} of 0.3 and 0.2 for simulated and real data respectively, was better able to estimate the number of MUPTs represented in a set of detected MUPs than the previous system, with average E_{NMUPTs} of 2.2 and 0.8 for simulated and real data respectively. For both the simulated and real data used, variations in A_c, A_r, and E_{NMUPTs} for the newly developed system were lower than for the previous system, which demonstrates that the new system can successfully adjust the assignment criteria based on the characteristics of a given signal to achieve robust performance across a wide variety of EMG signals, which is of paramount importance for successfully promoting the clinical application of EMG signal decomposition techniques.**

***Index Terms*—Clustering, electromyographic (EMG) signal decomposition, motor unit potential train validity, supervised classification.**

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I. Introduction

ELECTROMYOGRAPHIC (EMG) signal decomposition is the process of resolving an EMG signal into its component motor unit potential trains (MUPTs). The purpose of EMG signal decomposition is to provide an estimate of the firing pattern and motor unit potential (MUP) template and MUP shape stability of each active motor unit (MU) that contributed significant MUPs to a composite EMG signal. After preprocessing an acquired signal to remove background noise and low-frequency information, to sharpen the MUPs, and to increase discrimination between the MUPs created by different MUs, signal segments containing possible MUPs are detected. The detected MUPs, each represented by a vector of feature values, are then grouped into several MUPTs using clustering and/or supervised classification techniques. The MU firing patterns, MUP templates and MUP shape stabilities extracted from the obtained MUPTs can assist with the diagnosis of neuromuscular disorders [1]–[5], the understanding of motor control [6]–[8], and the characterization of MU architecture [9].

Numerous automatic and semi-automatic EMG signal decomposition methods have been proposed [10]–[37]. In fact, various MUP features, ranging from simply the MUP time samples to their wavelet coefficients, and different clustering and supervised classification techniques, ranging from simple minimum distance-based methods to classifier fusion techniques, have been employed to create an accurate decomposition algorithm. A recent comprehensive review of the algorithms developed for the decomposition of intramuscular EMG signals can be found in [11].

Existing EMG signal decomposition methods have been shown to be able to successfully decompose both the simulated and real EMG signals used for their evaluation [10]–[37]. However, the results obtained using the majority of these methods still depend on several factors such as the parameters used by the algorithms, the decomposability of the signal, and the MUP shape and MU firing pattern variability over the entire signal. The accuracy of an EMG signal decomposition depends on the validity of the MUPTs obtained by the decomposition algorithm which in turn depends on the criteria and parameters used to merge or split MUPTs. Unfortunately, there is no *a priori* known set of values for these parameters (e.g., assignment threshold) which is optimal for all EMG signals. These parameters and criteria have to be "tuned" by the user for the overall system to give the desired level of decomposition performance. Moreover, it can be shown both practically and conceptually that a better performance will be achieved when the assignment threshold is defined and then tuned for

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each MUPT individually, instead of using one assignment threshold value for all trains. Manually tuning the best value for assignment threshold for each train during decomposition is infeasible. In this paper, starting with the algorithms developed for the decomposition-based quantitative EMG (DQEMG) system [29], a knowledge-based framework has been used to develop a validity-based EMG decomposition (VBEMGD) system for addressing some of these issues.

The paper is organized as follows. Section II presents an overview of the supervised classifiers used to determine the validity of a MUPT. Details of the VBEMGD algorithms are presented in Section III followed by evaluation, results and discussion, and conclusion sections.

II. Validating a MUPT

A valid MUPT represents the activity of a single MU and may be contaminated by a small percentage of false-classification errors (FCEs). An invalid MUPT, on the other hand, either represents the activity of more than one MU (i.e., it is a merged MUPT) or contains an unacceptable percentage of FCEs (i.e., it is a contaminated MUPT). The class label of an extracted MUPT (single, merged, or contaminated) can be determined using the MU firing pattern and MUP shape information provided by the train. The classification, in general, is conducted by either an expert operator or by employing pattern recognition techniques. In this work, the validity of each MUPT obtained during decomposition is determined by two supervised classifiers: 1) the single/merged classifier that determines whether a given MUPT represents the activity of a single MU or not; and 2) the single/contaminated classifier that determines whether a train is a single train or a contaminated train. Below is a brief description of each classifier, details can be found elsewhere [38] and [39].

The single/merged classifier is a multi-classifier algorithm that determines the category of a given train (single or merged) by aggregating the classification results provided by two base classifiers: a MU firing pattern classifier and a MUP shape classifier. The MU firing pattern classifier employs ten features extracted from the inter-discharge interval (IDI) histogram and the instantaneous firing rate of the MU related to the given MUPT to determine the MU firing pattern based category of the train (i.e., single or merged). Detailed definitions and calculation methods for these features are presented elsewhere [38], [39]. In short, the majority of these features target the left side of the IDI distribution of the given MUPT, where short IDIs (i.e., the errors of interest) are reflected. The variation in the firing rate is represented by the mean consecutive difference of the instantaneous firing rate smoothed by a Hamming window of length 11. The MU firing pattern classifier is a support vector machine (SVM) classifier which uses a Gaussian radial basis function as a kernel. Details of the training and testing of the MU firing pattern classifier are given in [38], [39].

The MUP shape classifier assesses the similarity of the shapes of the MUPs assigned to it. Assuming MUPs generated by a single MU are similar in shape (but with possibly different degrees of similarity across different MUs), a MUPT is considered single (valid) if its MUPs have similar shapes. A SVM classifier and six features extracted from the MUPs of a train using cluster analysis methods as well as data mining techniques are used to determine the shape-based category (single/merged) of the train. Detailed definitions and calculation methods for these six features along with details related to the training and testing of the SVM used as the MUP shape classifier are given in [38]. In short, these features are defined and extracted based on the idea that the increase in the similarity of the MUP shapes of the two sub-trains created when splitting a MUPT is greater for an invalid MUPT than for a valid MUPT. In addition, the similarity between the MUP templates of the two sub-trains created by splitting an invalid train is less than that for a valid MUPT. The SVM used as the MUP shape classifier uses a Gaussian radial basis function as a kernel, see [38] for details related to the development of the MUP shape classifier.

The output of the MU firing pattern classifier and MUP shape classifier are aggregated using either an "AND" operation or a logistic regression classifier (LRC). When the "AND" logic is used for aggregation, the train under question is labeled single if it is classified as a single train by both the MU firing pattern classifier and MUP shape classifier. The LRC, however, determines MUPT validity by weighting the class-supports (i.e., support for being a valid train) provided by the MU firing pattern classifier and the MUP shape classifier. The "AND" aggregation method is more accurate than the LRC in detecting highly sparse merged MUPTs, but it is less accurate than the LRC in correctly classifying single MUPTs [38]. Therefore, during the early stages of an EMG signal decomposition, when the missed-classification error (MCE) rates (percentage of MCEs in a train) in the extracted MUPTs are high, the output of the MU firing pattern classifier and MUP shape classifier are aggregated using the "AND" logical. However, during the latter stages of decomposition, when MCE rates in the trains are low ($<65\%$), the LRC is used for aggregation.

The single/contaminated classifier assesses the MU firing patterns provided by a single MUPT to determine whether the percentage of FCEs in the train is acceptable ($<5\%$) or not. This classifier uses the same features used by the MU firing pattern classifier. However, a Fisher linear discriminate analysis based classifier and a data set that contained MUPTs with up to 20% FCE rate were used for training and testing [39].

III. EMG Decomposition Using MUPT Validity

The VBEMGD system decomposes a detected EMG signal off-line. VBEMGD consists of four major steps: signal preprocessing, MUP detection, and clustering and supervised classification of the detected MUPs. Following is a description of these steps.

A. Signal Preprocessing, MUP Detection, and Clustering

Signal preprocessing, MUP detection, and clustering of detected MUPs are completed using methods similar to those in DQEMG [29]. The given EMG signal is band-pass filtered using a low-pass difference filter [35] and the positions of suitable MUPs in the filtered signal are detected using a threshold crossing technique. For clustering and supervised classification, each detected MUP is represented using 2.56 ms of filtered data samples (i.e., 80 samples at 31250 Hz sampling rate), centered

about its peak value (i.e., about the position of maximum slope of the unfiltered MUP data).

Detected MUPs are clustered to obtain the initial information required for supervised classification such as estimates of the number of MUPTs, their MUP templates, and their MU firing pattern statistics. To extract such information, the MUPs detected in a specified portion (5 s duration with the highest intensity of MUPs) of the EMG signal are input to a shape and temporal-based clustering (STBC) algorithm [31] which groups the detected MUPs into several MUPTs using both firing time and shape information across multiple iterations (see [29] and [31] for more details).

B. Supervised Classification of Detected MUPs

Having the initial information about possible MUPTs provided by the clustering step, the detected MUPs are assigned to MUPTs using a supervised classifier. The objective here is to assign each MUP to the MUPT for which the MUP's time of occurrence and shape are more consistent with respect to the MU firing times and MUP shapes of the selected MUPT, respectively, than to the other MUPTs. Each of the MUPTs should have low MCE and FCE rates and represent the activity of a single MU that contributed detected MUPs to the given EMG signal. In this work, a new adaptive certainty-based classifier, which is partially based on a certainty-based classification algorithm previously developed for MUP classification [30], was developed for this purpose.

The adaptive certainty-based classification algorithm combines both MUP shape and MU firing pattern information to calculate the confidence of assigning a candidate MUP to a MUPT. The candidate MUP is assigned to the MUPT that has the greatest certainty value, if this value is greater than a certainty assignment threshold (C_{AT}). Otherwise, the MUP is left unassigned.

The certainties for assigning a candidate MUP are evaluated for the two trains that have the most and the next most similar MUP templates. These two trains are found by calculating the Euclidian distance between the candidate MUP and the MUP template of each MUPT. The certainties are calculated using MUP shape and MU firing pattern information. MUP shape certainty includes normalized absolute shape certainty (C_{ND}) and relative shape certainty (C_{RD}). The first represents the distance from the candidate MUP to the template of a train, normalized by the energy of the template. The second reflects the distance from the MUP to the most similar MUP template relative to the distance of the MUP to the next most similar MUP template. Assuming m_{j} represents the feature vector of the $\mathrm{MUP_j}$ being classified and, s_1 and s_2 respectively represent the feature vectors of the closest and the next closest MUP template to $\mathrm{MUP_j}$, $C^{\mathrm{j}}_{\mathrm{ND}}$ and $C^{\mathrm{j}}_{\mathrm{RD}}$ are evaluated by

$$C^{\mathrm{j}}_{\mathrm{ND\ i}} = max\left\{1 - \frac{\mathrm{r_i}}{\|\mathrm{s_i}\|^2}, 0\right\}; \mathrm{i} = 1, 2 \tag{1}$$

$$C^{\mathrm{j}}_{\mathrm{RD\ i}} = 2 - \mathrm{i} + (-1)^{\mathrm{i}}\frac{\mathrm{r_1}}{2\mathrm{r_2}}; \mathrm{i} = 1, 2 \tag{2}$$

where r_{i} is the Euclidean squared distance between s_{i} and m_{j}.

Firing pattern certainty, C_{FC}, measures the consistency of the occurrence time of $\mathrm{MUP_j}$ relative to the established MU firing pattern of a MUPT. Denoting t_{j} as the occurrence time of the given MUP (i.e., $\mathrm{MUP_j}$) and t_{pi} and t_{ni} as the occurrence times of the previous and next MUPs in $\mathrm{MUPT_i}$, the firing pattern certainty of assigning $\mathrm{MUP_j}$ to MUPT_i $(C^{\mathrm{j}}_{\mathrm{FCi}})$ is given by

$$C^{\mathrm{j}}_{\mathrm{FCi}} = C^{\mathrm{j}}_{\mathrm{pi}}\left(|\mathrm{t_j} - \mathrm{t_{pi}}|\right) \times C^{\mathrm{j}}_{\mathrm{ni}}\left(|\mathrm{t_j} - \mathrm{t_{ni}}|\right) \tag{3}$$

where

$$C^{\mathrm{j}}_{\mathrm{pi}}\left(|\mathrm{t_j} - \mathrm{t_{pi}}|\right) = \sum_{\mathrm{n}=1}^{40} \frac{1}{\sqrt{\mathrm{n}}} exp\left(-\frac{\left(|\mathrm{t_j} - \mathrm{t_{pi}}| - \mathrm{n}\mu_{\mathrm{i}}\right)^2}{2\mathrm{n}\sigma_{\mathrm{i}}^2}\right). \tag{4}$$

The two parameters μ_{i} and σ_{i} represent the mean and standard deviation of the IDIs of $\mathrm{MUPT_i}$ estimated using the EFE algorithm [40]. If μ_{i} and σ_{i} are not available or $|\mathrm{t_j} - \mathrm{t_{pi}}| > \mu_{\mathrm{i}} + 3\sigma_{\mathrm{i}}$, $C^{\mathrm{j}}_{\mathrm{pi}}(|\mathrm{t_j} - \mathrm{t_{pi}}|)$ is set equal to 0.2. The value for $C^{\mathrm{j}}_{\mathrm{ni}}(|\mathrm{t_j} - \mathrm{t_{ni}}|)$ is estimated as $C^{\mathrm{j}}_{\mathrm{pi}}(|\mathrm{t_j} - \mathrm{t_{pi}}|)$ by replacing t_{pi} with t_{ni} in (4) [41].

Having the shape certainties and the firing pattern certainty, the overall certainties for assigning $\mathrm{MUP_j}$ to one of the two selected MUPTs are estimated as

$$C^{\mathrm{j}}_{\mathrm{i}} = C^{\mathrm{j}}_{\mathrm{NDi}} \times C^{\mathrm{j}}_{\mathrm{RDi}} \times C^{\mathrm{j}}_{\mathrm{FCi}}; i = 1, 2. \tag{5}$$

If $max(C^{\mathrm{j}}_1, C^{\mathrm{j}}_2) > C_{\mathrm{AT}}$, $\mathrm{MUP_j}$ is assigned to the MUPT which has the greatest certainty value, otherwise it is left unassigned.

In order to accommodate nonstationarity in MUP shapes, the algorithm updates the MUP templates with each MUP assignment. The MUP templates are calculated using a moving average for which the weights are the certainties with which MUPs are assigned to the MUPTs. If $\mathrm{MUP_j}$ is assigned to $\mathrm{MUPT_i}$ with certainty $C^{\mathrm{j}}_{\mathrm{i}}$ higher than the updating threshold (0.6 in this work) the template of $\mathrm{MUPT_i}$ (s_{i}) is updated as

$$\mathrm{s}^{\mathrm{New}}_{\mathrm{i}} = \frac{\mathrm{s_i} + C^{\mathrm{j}}_{\mathrm{i}} \times \mathrm{m_j}}{1 + C^{\mathrm{j}}_{\mathrm{i}}} \tag{6}$$

where m_{j} is the feature vector of $\mathrm{MUP_j}$.

Once each classification-pass through the set of detected MUPs is completed and before decomposition continues, the validity of each extracted MUPT is assessed as described in Section II. Invalid trains are detected, corrected and have their C_{AT} adjusted. Trains that are classified as merged by the single/merged classifier are split into valid trains; the single MUPTs that were classified as contaminated by the single/contaminated classifier have their FCEs corrected using an automated MUPT editing algorithm [42].

The parameter C_{AT} has a high impact on the level of MCE and FCE rates in the resulting MUPTs. Unfortunately, there is no certain value for C_{AT} which is appropriate for all EMG signals. Moreover, it can be shown both practically and conceptually that a better performance will be achieved when a C_{AT} value is defined and then tuned for each MUPT individually, instead of using one C_{AT} for all trains. Manually tuning the best C_{AT} value for each train during decomposition is infeasible. In VBEMGD, the C_{AT} value for each MUPT is adjusted based on its validity (i.e., an adaptive adjustment of the assignment threshold). Merged invalid MUPTs are split and the C_{AT} of contaminated MUPTs is increased by a step of Δ. The C_{AT} for valid

trains is reduced by an amount calculated using the pseudo-correlation (PsC) [23] between the MUP template of this train and that of the closest train as follows:

$$C_{\mathrm{AT}}^{\mathrm{NEW}} = \Delta + [max(0.02, C_{\mathrm{AT}}) - \Delta] \times exp\left(-\frac{N_{\mathrm{itr}}}{\eta}\right) \quad (7)$$

where N_{itr} is the number of iterations that the classification algorithm has passed through in assigning all of the MUPs and η is the decreasing rate of the C_{AT} for a valid MUPT and is given by

$$\eta = \begin{cases} \frac{1}{3} & \mathrm{PsC} \leq 0.1 \\ 1 & 0.1 < \mathrm{PsC} \leq 0.3 \\ 2 & 0.3 < \mathrm{PsC} \leq 0.5 \\ 4 & \mathrm{PsC} > 0.5 \end{cases} \quad (8)$$

The C_{AT} value of a MUPT is not decreased or increased below 0.005 or above 0.99, respectively.

In addition to splitting or editing invalid MUPTs, the chance of merging single MUPTs is evaluated. Pairs of MUPTs that have similar MUP templates are merged if the resulting train is valid. MUP template similarity is measured by estimating the PsC between the MUP templates of the two selected MUPTs.

The MU firing pattern statistics of each MUPT are updated after each pass. The MUP assignment and MUPT splitting, editing, and merging steps are repeated until either, the maximum number of iterations is exceeded or the MUPTs are stable. If trains are merged or split at least one more supervised classification pass will be completed.

A flowchart of the VBEMGD system is presented in Fig. 1. The adaptive nature of the system, which adjusts the C_{AT} value for each MUPT based on the estimated validity of the train, leads to improved decomposition results in terms of estimating the correct number of MUPTs represented in the set of detected MUPs and decreasing MCE and FCE rates in each MUPT.

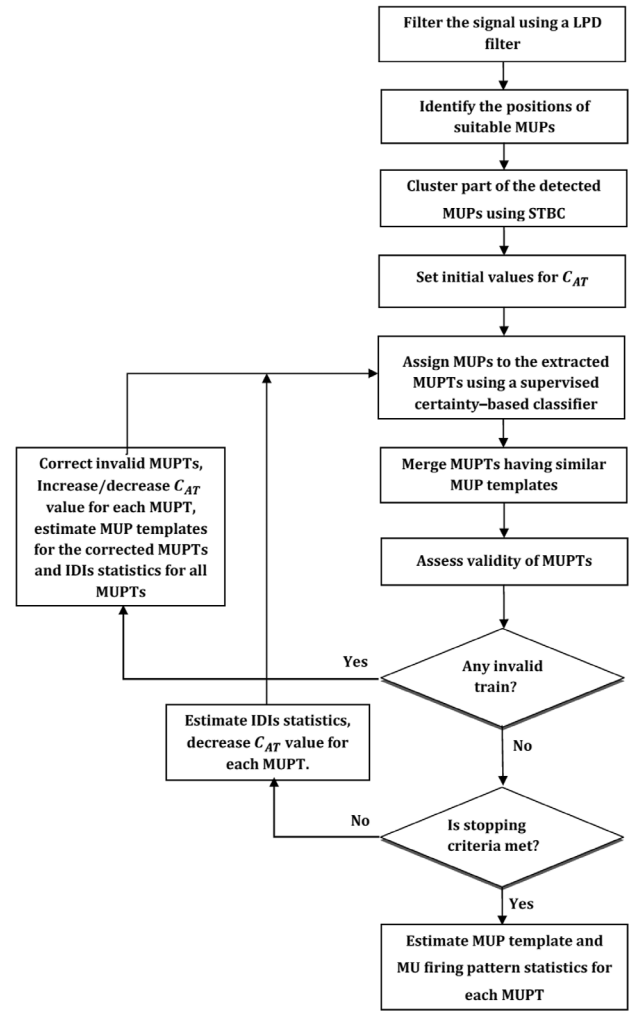

Fig. 1. Flowchart of the validity-based EMG decomposition system.

IV. Evaluation Methodology

The performance of VBEMGD was evaluated using several simulated and real EMG signals. For each EMG signal, the MU discharge patterns provided either by the EMG signal simulator employed or by a human expert operator were used as reference. Following are details of each data set.

For simulated data, 32 EMG signals were generated using a physiologically-based EMG signal simulation algorithm [43]. This EMG signal generation tool creates simulated intramuscular EMG signals with different complexities such as different numbers of active MUs, different degrees of MUP shape and/or IDI variability (represented by the amount of MUP jitter and IDI CV), and different signal intensities [represented by the average number of MUPs per second (pps)]. Using these algorithms, EMG signals similar to those detected from normal, myopathic and neurogenic muscles, with different degrees of involvement, can be simulated.

For real data, several single-channel EMG signals provided by McGill [44]; and multi-channel EMG signals provided by Florestal, Mathieu, and McGill [14] were used. The single-channel signals were recorded using a monopolar needle electrode during low-level isometric contractions of the brachial biceps muscles of normal subjects. The multi-channel signals were recorded simultaneously using three or four pairs of fine wire electrodes from the brachioradialis muscles of three normal subjects during low-level isometric contractions. Of the MUs contributed to each signal, only the discharge patterns of those MUs that were selected by the expert as accurately identified patterns were considered as reference and used for evaluation. For this data set, the signals detected by each electrode were considered as single-channel EMG signals. It should be pointed out that the detection threshold used by the EMGLAB software [25] to manually decompose these real signals are lower than the threshold used by both DQEMG and VBEMGD.

Four indexes were used for evaluation: assignment rate (A_{r}), accuracy (A_{c}), correct classification rate (CC_{r}), and error in finding the correct number of MUPTs (E_{NMUPTs}). These four indexes are given by

$$A_{\mathrm{r}}\% = \frac{\text{Number of MUPs assigned}}{\text{Number of MUPs detected}} \times 100 \quad (9)$$

$$A_{\mathrm{c}}\% = \frac{\text{Number of MUPs correctly classified}}{\text{Number of MUPs assigned}} \times 100 \quad (10)$$

$$CC_{\mathrm{r}}\% = \frac{\text{Number of MUPs correctly classified}}{\text{Number of MUPs detected}} \times 100 \quad (11)$$

TABLE I
CHARACTERISTICS OF THE SIMULATED EMG SIGNALS USED ALONG WITH THE PERFORMANCE OF THE DECOMPOSITION ALGORITHMS OF THE DQEMG AND THE VALIDITY-BASED DECOMPOSITION SYSTEM FOR THESE SIGNALS. THE SIGNALS WERE GROUPED BASED ON THEIR INTENSITY. FOR EACH GROUP OF EMG SIGNALS USED, THE RANGE OF INTENSITY, NUMBER OF CONSTITUENT MUPTS, MDI, ALONG WITH MEAN $[min, Max]$ OF THE ACCURACY INDEX USED ARE REPORTED. NOTE: FOR ENMUPTS THE MEAN ABSOLUTE VALUE IS PRESENTED

				Original DQEMG				VBEMGD			
Signals	**Intensity (pps)**	**No. of MUPTs**	**MDI**	A_r (%)	A_c (%)	CC_r (%)	E_{NMUPTs}	A_r (%)	A_c (%)	CC_r (%)	E_{NMUPTs}
1–5	20–55	3–6	1.3–3.0	91.0 [88.8,92.6]	97.0[93.1,99.4]	88.3[82.7,92.9]	0.0[0,0]	97.9[97.4,98.6]	98.3[96.4,100]	96.2[94.3,98.6]	0.0[0,0]]
6–15	55–90	6–9	0.3–1.8	83.2 [70.7,98.1]	85.4 [68.8,96.9]	71.0 [52.6,87.0]	1.5[–1,4]	92.5[87.7,95.8]	96.4[88.6,99.6]	89.2[79.9,95.4]	0.1[0,1]]
16–28	90–125	7–11	0.3–0.9	80.9[71.9,87.9]	82.6[70.2,90.3]	66.8[54.5,78.2]	2.8[1,5]	90.2[86.3,94.3]	92.4[85.8,97.6]	83.4[75.5,89.1]	0.4[0,1]]
29–32	125–160	10–11	0.3–1.1	72.3[62.4,81.6]	83.9 [80.2,89.5]	60.8 [51.3,68.8]	5.3[4,7]	86.3[87.6,90.7]	89.3[84.6,94.7]	77.2 [66.9,83.0]	1.0[0,2]]
		Mean		**82.3**	**86.7**	**71.6**	**2.3**	**91.6**	**94.2**	**86.4**	**0.3**
		STD		**8.6**	**8.3**	**11.5**	**2.0**	**4.2**	**4.6**	**7.5**	**0.5**

TABLE II
PERFORMANCE OF THE ORIGINAL DECOMPOSITION ALGORITHMS OF THE DQEMG SYSTEM AND THE VALIDITY-BASED DECOMPOSITION SYSTEM FOR REAL EMG SIGNALS. NOTE: FOR E_{NMUPTs} THE MEAN ABSOLUTE VALUE IS PRESENTED. FOR EACH GROUP OF EMG SIGNALS USED, THE RANGE OF INTENSITY, NUMBER OF CONSTITUENT MUPTS, MDI, ALONG WITH MEAN $[min, Max]$ OF THE ACCURACY INDEX USED ARE REPORTED

				Original DQEMG				VBEMGD			
Signals	**Intensity (pps)**	**No. of MUPTs**	**MDI**	A_r (%)	A_c (%)	CC_r (%)	E_{NMUPTs}	A_r (%)	A_c (%)	CC_r (%)	E_{NMUPTs}
1–3	20–55	3–5	30.0–32.0	97.7 [93.0,100.0]	99.9[99.9,100.0]	97.6[92.9,100.0]	0.0[0,0]	99.2 [97.9,100.0]	99.9 [99.8,100.0]	99.2 [97.9,100.0]	0.0[0,0]]
4–15	55–90	6–12	1.2–31.3	96.1[84.7,99.8]	95.4 [61.7,100.0]	91.6 [61.2,99,7]	0.6[–5,1]	98.8 [96.3,100.0]	99.5 [98.5,100.0]	98.3 [94.9,100.0]	0.1[0,1]]
16–25	90–125	9–13	0.7–21.2	92.3[82.7,96.8]	95.0[72.3,99.9]	87.6[69.7,96.7]	1.1[–3,4]	96.1[91.0,98.9]	98.8[95.4,99.7]	94.9 [86.7,98.6]	0.3[0,2]]
26–30	125–160	13–15	1.1–7.1	88.9[86.1,91.4]	95.3[89.9,98.7]	84.7 [77.8,88.9]	1.0[–1,1]	94.6[93.0,96.6]	98.6 [98.3,98.9]	93.3[91.5,95.5]	0.2[0,1]]
		Mean		**93.8**	**95.7**	**89.7**	**0.8**	**97.2**	**99.1**	**96.4**	**0.2**
		STD		**5.0**	**8.7**	**9.3**	**1.3**	**2.4**	**0.9**	**3.1**	**0.5**

$$E_{\mathrm{NMUPTs}} = \text{Number of extracted MUPTs} - \text{Number of expected MUPTs} \tag{12}$$

where the number of expected MUPTs equals to the number of MUPTs identified by a human expert or included by the simulator which had a MUP template with second derivative values greater than 500 $\mu\mathrm{V/s}^2$. All extracted MUPTs were matched with the expected reference trains and performance indexes were calculated using the methods described in [45].

In this work, because superimposed MUPs are not resolved, not all detected MUPs are assigned. As such, A_r reflects the percentage of detected MUPs assigned, while A_c represents the accuracy (or precision) with which MUP assignments are made. CC_r represents the net percentage of correctly classified MUPs or sensitivity with which MUP assignments are made. CC_r is dependent on the number of superimposed MUPs (i.e., the intensity of the EMG signal being decomposed) as well as the proficiency of the decomposition algorithm used.

The ease with which MUPs can be correctly assigned to a MUPT is different within and across signals. To quantify this concept and to better assess decomposition performance a MUPT decomposability index (DI) was defined and used

$$\mathrm{DI_i} = \underset{\mathrm{j=1,2,\ldots,m;j\neq i}}{min}\left(\frac{\mathrm{SB_{i,j}}}{\mathrm{SW_{i,j}}}\right) \tag{13}$$

where m is the number of MUPTs of a given EMG signal, $SB_{\mathrm{i,j}}$ is the between train variance of the ith and jth MUPT, respectively, and $SW_{\mathrm{i,j}}$ is the within train variance estimated for these two trains. The variances were estimated for MUP shapes. The DI is conceptually similar to the Fisher criterion used to estimate the separability between two clusters [46]. For each EMG signal, the median of the DIs of its MUPTs (MDI) were estimated and represent as a measure of the decomposability of the signal.

V. RESULTS AND DISCUSSION

Performance results for VBEMGD and that of the original decomposition algorithms of DQEMG for both simulated and real data are summarized in Tables I and II, respectively. These results were produced using these two experimentally determined parameters: $C_{\mathrm{AT}} = 0.02$ for DQEMG and $\Delta = 0.005$ for VBEMGD. For each data set, the performance for each group of signals used along with the mean and standard deviation (STD) for the performance indexes over all signals is reported. Statistical comparisons of the average values were conducted using paired t-tests ($\alpha = 0.05$), while comparisons of the STD values were conducted using F-tests ($\alpha = 0.05$).

For the simulated data set (i.e., Table I), VBEMGD has significantly improved decomposition results in terms of all four performance indexes ($p < 0.00001$), with an average increase in CC_r of 14.8%. In addition, VBEMGD has lower STD for all performance measures ($p < 0.02$), which shows that VBEMGD has better overall and less variable performance. The improvement in decomposition results (especially CC_r and E_{NMUPTs}) increases as the complexity of the signal increases, such that for the last four signals, each having intensity >125 pps, the CC_r values are improved on average by 16.4%.

In general, the results obtained using the real data presented in Table II are consistent with those presented for the simulated data; VBEMGD outperformed DQEMG in terms of the average and STD values of all four performance measures, with an average CC_r increase of 6.7%. This shows that VBEMGD has

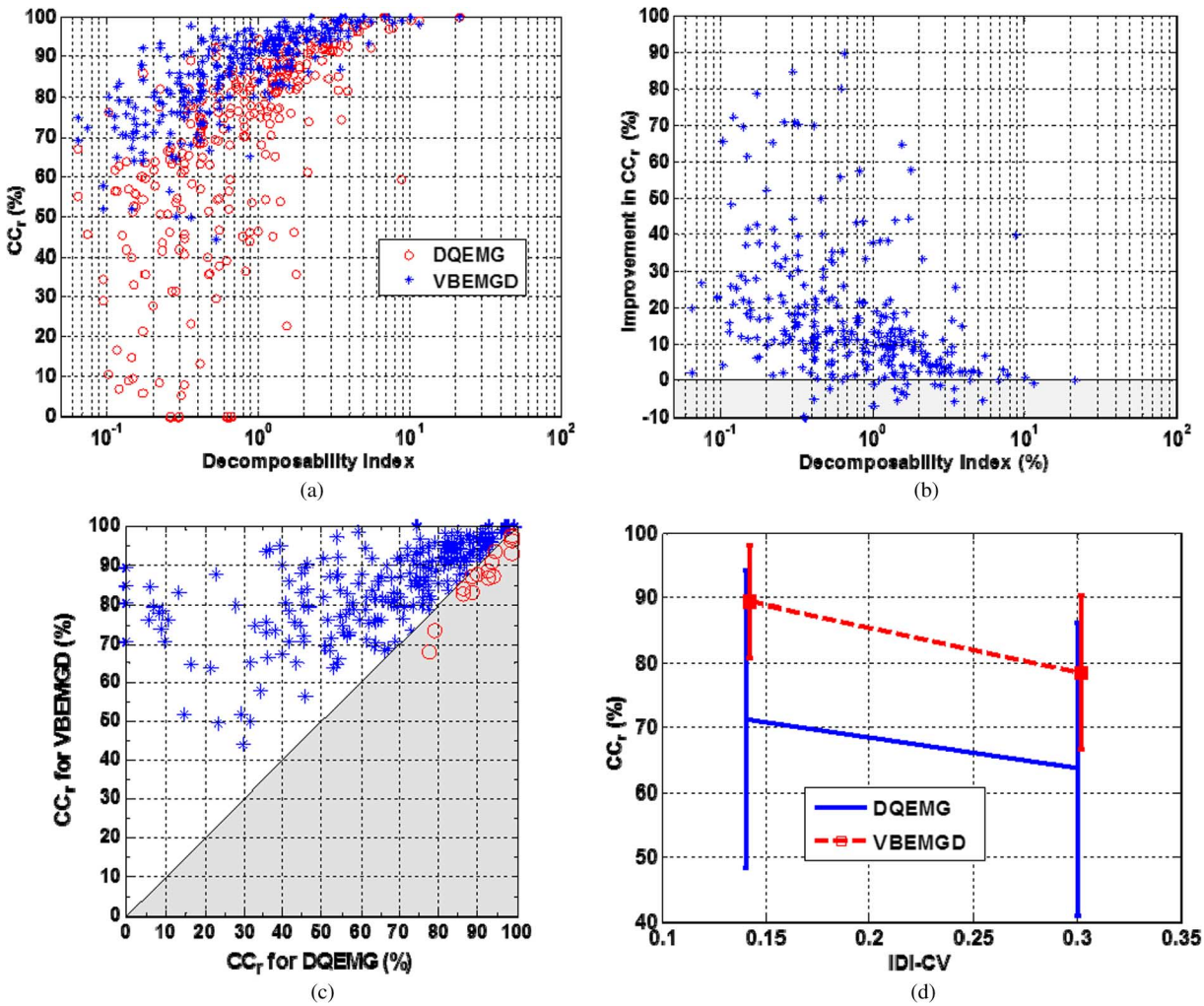

Fig. 2. Comparison of the performance of the DQEMG and VBEMGD for the MUPTs in the simulated data set. (a) CC_{r} for DQEMG (*) and VBEMGD (o) versus decomposability index. (b) Improvement in CC_{r} of the MUPTs versus their decomposability index. For those MUPTs that improvement is <0, the CC_{r} decreased when VBEMG was employed for decomposition. (c) CC_{r} of the VBEMGD versus that of the DQEMG. The asterisks indicate MUPTs that VBEMGD increased CC_{r} while circles represent those that their CC_{r} decreased. (d) CC_{r} with respect to firing pattern variability.

higher overall and less variable (or more robust) performance for real signals as well. However, both systems performed better on the real data set than on the simulated data set. One possible reason is that the MU firing pattern and MUP shape variability in the real signals are lower than that in the simulated signals. The other possible reason is the higher decomposability of MUPTs in the real data relative to those in the simulated data.

An interesting and significant advancement of VBEMGD over DQEMG is in estimating the number of MUPTs comprising the EMG signals used. As shown in both Tables I and II for most EMG signals VBEMGD correctly estimated the number of trains in the signals; however, DQEMG extracted up to seven extra MUPTs for some signals. Moreover, VBEMGD either provided the true number of MUPTs or extra trains but never underestimated this parameter; on the other hand, with DQEMG up to 5 MUPTs were missed for some signals (e.g., signal #23 in Table II). In addition, the increases in A_{r} without a drop or with an improvement in A_{c} seen in Tables I and II, indicate that VBEMGD can correctly lower the assignment thresholds of individual trains without affecting assignment accuracies and thus improve overall performance.

Fig. 2(a)–(d) further emphasize the advantages of VBEMGD over DQEMG. As shown in Fig. 2(a), for the majority of the difficult to decompose MUPTs (i.e., MUPTs with DI < 2) the CC_{r} values obtained using DQEMG are <60% while the CC_{r} values provided by VBEMGD for these trains are >60%. Such improvement is better illustrated in Fig. 2(b) where it can be seen that the improvement in CC_{r} achieved when using VBEMGD is considerable for hard to decompose MUPTs such that for some MUPTs the improvement is even >50%.

Fig. 2(c) compares the CC_{r} achieved using DQEMG for each MUPT with that obtained by VBEMGD. As shown, with VBEMGD the CC_{r} values for the majority of the MUPTs were increased, but for a few MUPTs the CC_{r} values were decreased. Nevertheless, the decreases in CC_{r} values were small as most of the circles (indicating these cases) are along the diagonal.

Increases in MU firing pattern or MUP shape variability can decrease the performance of a decomposition system. Fig. 2(d) shows the effect of MU firing pattern variability (represented by IDI-CV) over the CC_{r} values for both VBEMGD and DQEMG. The advantage of VBEMGD over DQEMG in terms of im-

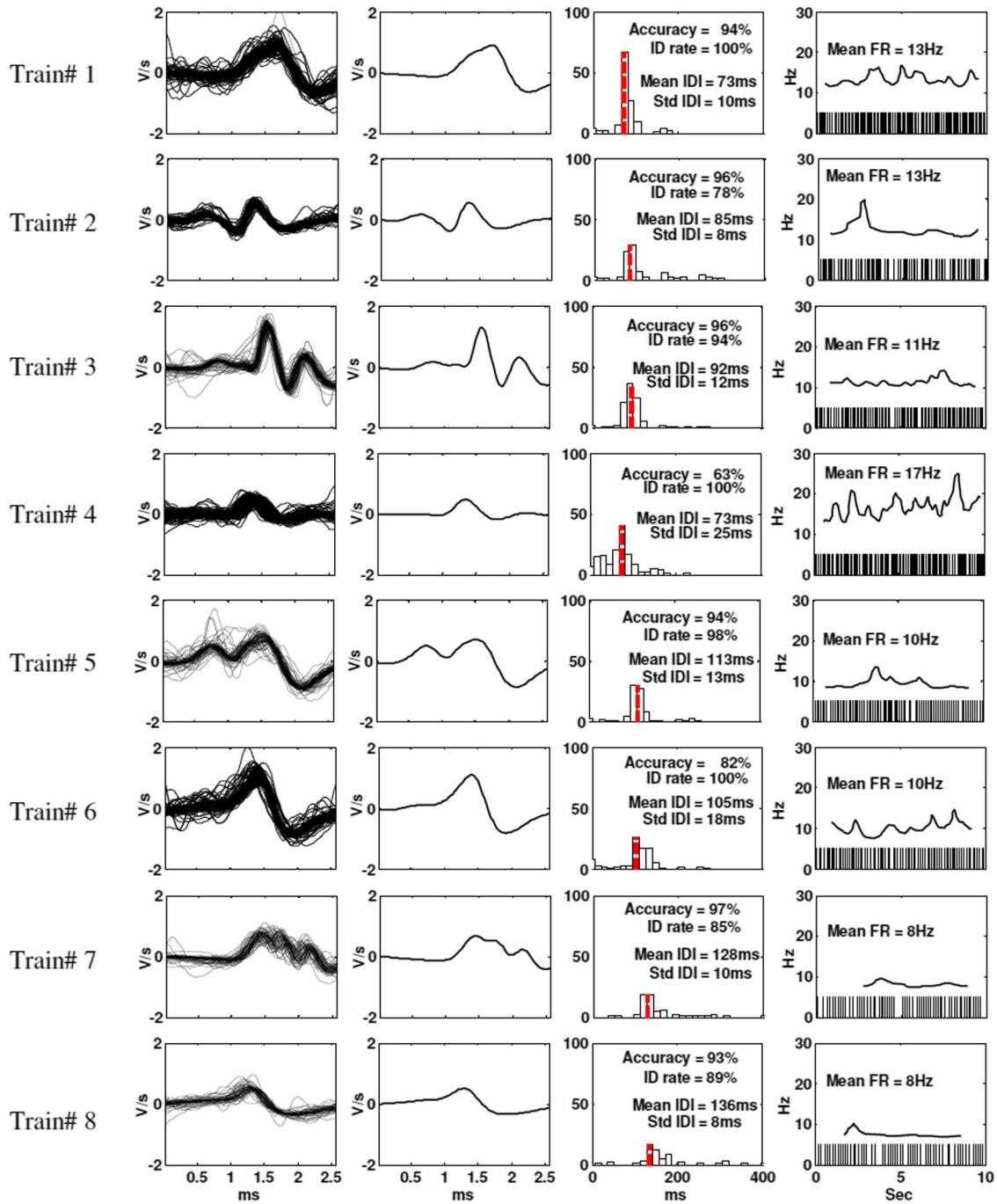

Fig. 3. Decomposition results for simulated EMG signal 14 obtained using the decomposition algorithms of the DQEMG. The first column shows the shimmer plot of the slope of the MUPs assigned to each MUPT. The second column shows the template for these MUPs. The third column presents the IDI histogram of each MU. Finally, the last column presents the firing pattern of each MU.

proving CC_{r} and its variability is also apparent in this figure. In short, for the EMG signals with relatively high jitter and IDI-CV values studied, the improvement in CC_{r} gained using VBEMGD was significant.

Figs. 3 and 4 present decomposition results for simulated EMG signal 14 using DQEMG and VBEMGD, respectively. The EMG signal used is composed of 9 MUPTs and was simulated using a jitter value of 50 μs and an IDI-CV of 0.15. For each MUPT, the accuracy and identification rate (ID rate) [39] are presented. Comparing MU firing patterns plots and the accuracy and identification rate values presented for each train in these two figures reveals the advantages of using VBEMGD. As shown, both the accuracy and completeness (ID rate) of the majority of the trains created by VBEMGD are improved relative to those created by DQEMG. Similar improvement can also be seen in the IDI histogram and MU firing rate plot of each MUPT. More importantly, DQEMG underestimated the expected number of MUPTs. Two MUPTs (Trains 4 and 9 according to Fig. 4) were mistakenly merged and reported as Train 4. On the other hand, VBEMGD correctly estimated the number

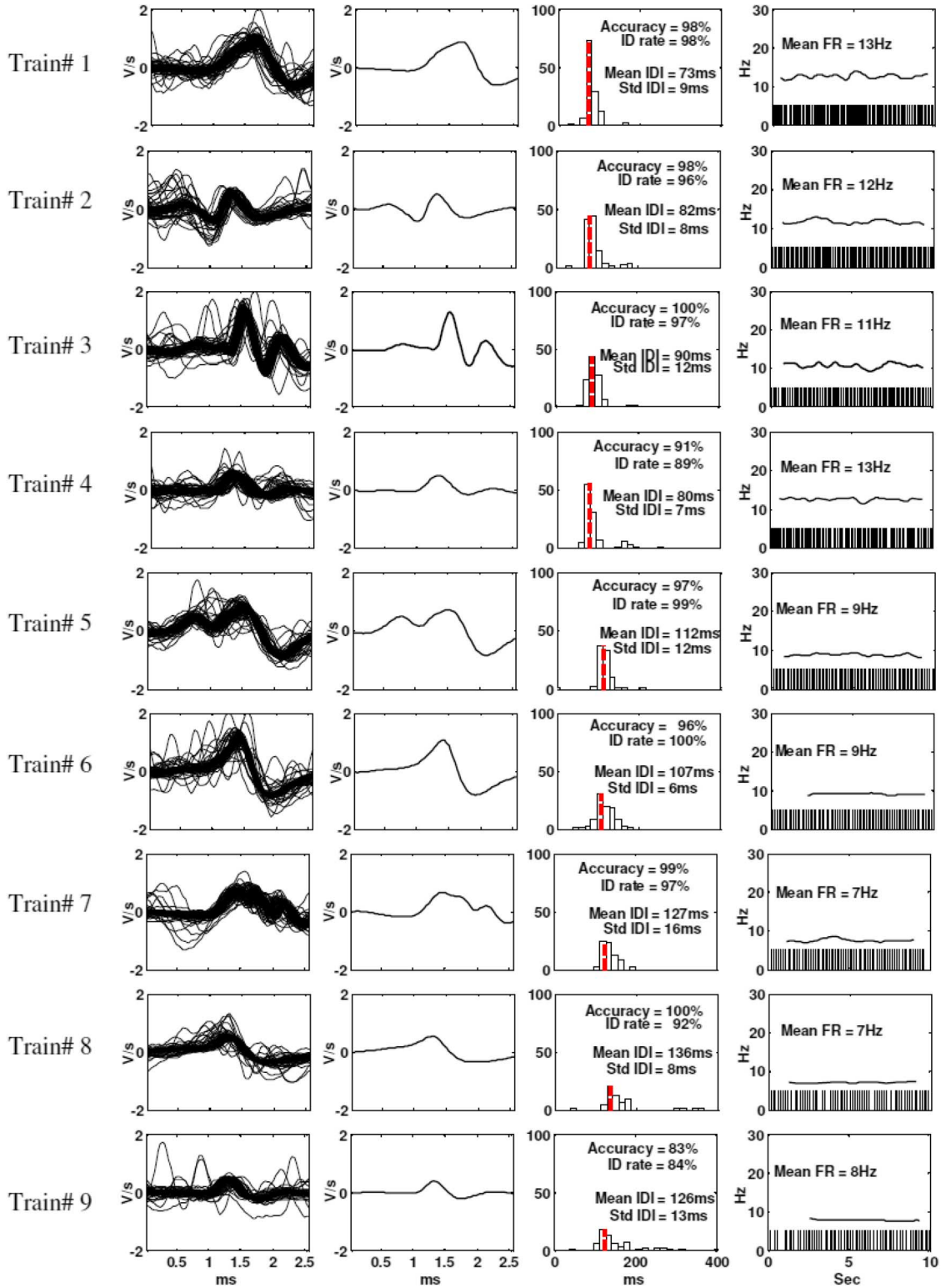

Fig. 4. Decomposition results for simulated EMG signal 14 obtained using VBEMGD. The first column shows the shimmer plot of the slope of the MUPs assigned to each MUPT. The second column shows the template for these MUPs. The third column presents the IDI histogram of each MU. Finally, the last column presents the firing pattern of each MU.

of expected MUPTs. These VBEMGD performance improvements can be clinically significant and may help promote the clinical application of VBEMGD.

Both VBEMGD and DQEMG do not consider low amplitude MUPs, which are composed of low frequency components and created by MUs with few muscle fibers close to the electrode detection surface. If such MUPs were detected and then considered for clustering and supervised classification, the accuracies of both systems may not be as high as those presented in Tables I and II. Finally, the accuracies of both DQEMG and VBEMGD

for EMG signals contaminated by high levels of noise may be lower than the values reported for the simulated and real EMG signals used in this work.

Both DQEMG and VBEMGD assume that the mean and standard deviation of the IDIs of the MUs that contributed to the signal being decomposed did not change significantly during signal detection. Such assumptions are valid for EMG signals detected clinically during short-term isotonic contraction; however, these assumptions may not be realistic for signals detected during either force-varying or long duration contractions. Such assumptions may limit the use of both DQEMG and VBEMGD for research applications where the decomposition of signals detected during non-isotonic or long duration contractions are required. Nevertheless, DQEMG has been useful for the decomposition of intramuscular EMG signals acquired for clinical applications [2], [3], [29], [46]–[48].

VBEMGD and DQEMG have also been applied to clinically-detected EMG signals. A challenge in decomposing clinically-detected EMG signals is that MUP shape variability and/or MU firing pattern variability can be higher than in simulated EMG signals and in EMG signals carefully acquired from control subjects in research labs. Such variability can be due to needle movement, patient compliance, and/or the affects of disease. A quantitative performance comparison was not completed because "gold standard" decomposition results for the clinically-detected EMG signals were not readily available. However, qualitative assessment of MUPTs, as used in [47]–[49], showed that VBEMGD performed better than DQEMG in decomposing clinical EMG signals especially in finding the correct number of MUPTs and reducing MCE rates.

Validating MUPTs during EMG signal decomposition and correcting invalid MUPTs takes time. The average processing time for DQEMG for the simulated data used was 15.9 s while for VBEMGD it was 23.6 s; (i.e., DQEMG is approximately 1.5 times faster than VBEMGD). The algorithms are currently encoded in Matlab (The Mathworks, Natick, MA) and processing was carried out on a computer with an Intel Core 2 Duo E8400 CPU (6 M Cache, 3.00 GHz clock, 1333 MHz FSB)and 3 GB of RAM. For both algorithms, the processing time is proportional to the number of MUPTs and the number of MUPs assigned. The average rate of increase of computation time versus intensity (pps) for DQEMG and VBEMGD was estimated to be 0.3 and 0.5 s/pps, respectively. Nonetheless, both DQEMG and VBEMGD are fast enough to be useful in clinical environments.

VI. Conclusion

An innovative system for EMG signal decomposition has been presented. The MUPs comprising a given EMG signal are first detected using a threshold-based algorithm and then grouped into several MUPTs using a shape and temporal-based clustering algorithm and an adaptive certainty-based classification algorithm. The developed system uses MUPT validity techniques to efficiently estimate the numbers of MUPTs comprising a given EMG signal by splitting merged MUPTs and merging MUPTs having similar MUP templates. To reduce the number of FCEs and MCEs in the extracted MUPTs: 1) contaminated MUPTs are identified and corrected; 2) the assignment threshold for each train is increased or decreased based on the estimated validity of the train. The rate of decrease is dependent on the similarity of MUP templates between trains (i.e., adaptive classification). Quantitative and qualitative evaluation using several simulated and real EMG signals demonstrate that VBEMGD is more robust and accurate but slower than the decomposition algorithms of DQEMG. This increased robustness and accuracy of VBEMGD will hopefully help promote more frequent application of clinical QEMG techniques.

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